L10: How do Dense Clusters Affect Cascades?

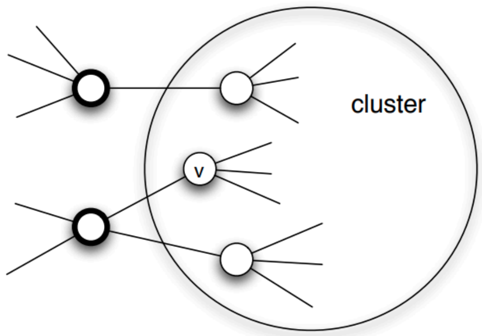

Image source: Textbook [***"Networks, Crowds and Markets" (https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf)***](https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf) *book, by Easley and Kleinberg, Chapter 19*

We will first prove the following result:

Suppose that we model the diffusion process using the Linear Threshold Model with threshold value θ.

If the set of initially active nodes is A, then the cascade will NOT cover the whole network if the network includes a cluster of initially inactive nodes with density greater than $1-\theta$.

Proof: We will prove this by contradiction - we start with an assumption, then following a series of steps which leads to a result that cannot be true. Consider such a cluster of initially inactive nodes with density greater than $1-\theta$. Assume the opposite of what we want to prove, i.e., that one or more nodes in this cluster eventually become active. Let *v* be the **first** such node in the cluster.

At the time *v* became active, its only active neighbors could be outside the cluster. But since the cluster has density greater than $1-\theta$, at least a fraction $1-\theta$ of *v*'s neighbors are in the cluster. So, less than a fraction θ of *v*'s neighbors are outside the cluster. Even if all of those neighbors were active, that would not be enough to activate *v*. This leads to a contradiction, and so it cannot be that the cluster eventually includes an active node.

Moreover, we can show the following related result:

If a network cascade that starts from a set of initial nodes A does not cover the whole network, then the network must include a cluster of density greater than $1-\theta$.

Food For Thought

Prove the last result stated above. The visualization at the right will help you to think about the proof.